

TOOL AND TROUBLESHOOTING

1. List three real-life scenarios where a multimeter would be used to troubleshoot a computer or smartphone issue, and explain in one line how the multimeter helps in each case

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Scenario	How the Multimeter Helps
Computer not turning on	Checks the power supply voltage to see if it is working properly.
Smartphone not charging	Checks the charging port and cable for proper voltage and continuity.
Motherboard problem	Checks voltage levels on the motherboard to find faulty power connections.

2. Watch a short YouTube video demo of a POST card in action (search for 'POST card motherboard test'), then write a brief summary (3-4 lines) describing what the POST card displays and what problem it helped identify in the video.
Hint: Focus on the codes shown and how they relate to hardware issues.

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I watched a YouTube demonstration of a POST card used for motherboard troubleshooting. The POST card displays **HEX codes** such as numbers and letters while the BIOS checks the computer hardware. The code shows which stage of the startup test the motherboard has reached. If the system stops at a particular code, it can help identify a **hardware or motherboard problem**. ([youtube.com](https://www.youtube.com/watch?v=8G6R1YUW0j0))
[YouTube – Beep Codes, POST Cards and PSU Testers](https://www.youtube.com/watch?v=8G6R1YUW0j0)

3. Download and install any free software diagnostic tool (such as HWMonitor, Speccy, or CrystalDiskInfo), run it on your laptop or desktop, and take a screenshot of the main dashboard showing system health details.
Constraint: Blur or crop out any personal information before submitting.

=> **Steps:**

1. Download and install **HWMonitor**.
2. Open HWMonitor on your laptop/desktop.
3. Check the main screen for **CPU temperature, voltage, fan speed, RAM, and other hardware details**.
4. Take a screenshot of the main dashboard.
5. **Crop or blur personal information** such as your username, computer name, serial number, or IP address.
6. Add the screenshot to your assignment.

4. Compare the use of a multimeter, POST card, and software diagnostics by creating a table with three columns (Tool, What it checks, Example Problem Detected) and fill in at least one example for each tool.

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Tool	What it checks	Example Problem Detected
Multimeter	Checks voltage, resistance, and continuity.	Faulty power supply or charging cable.
POST Card	Checks BIOS/POST codes during computer startup.	RAM, CPU, or motherboard hardware problem.
Software Diagnostics	Checks temperature, storage health, RAM, and system information.	CPU overheating or failing SSD/HDD.